Chromatography

Answer key with worked solutions

A. Understanding

1. Phases.

> (a) The **stationary phase** stays in place (paper, silica, a column coating); the **mobile phase** moves through or over it (a solvent or a gas). (b) Each substance is shared between the two phases according to its **attraction** to each. A substance held more strongly by the stationary phase spends more time there and **moves more slowly**, so the substances travel different distances and separate.

B. Rf values

2. Rf values.

> (a) $\frac{2.10\ \text{cm}}{8.40\ \text{cm}} = \mathbf{0.250}$; $\frac{4.62\ \text{cm}}{8.40\ \text{cm}} = \mathbf{0.550}$; $\frac{6.72\ \text{cm}}{8.40\ \text{cm}} = \mathbf{0.800}$
> (b) **X** (0.25) and **Y** (0.55). (c) The spot at Rf 0.800 matches **none** of the standards (Z is 0.70): it is a different, unidentified substance.

3. Technique.

> (a) Pencil (graphite) does not dissolve; **ink would separate** and add spots. (b) So the spots are **not washed into the solvent**. (c) To keep the air **saturated with solvent vapour**, so the solvent does not evaporate from the plate and the front moves evenly. (d) The front **evaporates quickly** and becomes invisible, and its distance is needed for Rf.

C. Polarity

4. Polarity.

> (a) **Hexane > propanone > ethanoic acid.** Silica is very polar. Non-polar hexane is held least and moves furthest; propanone is polar (C=O); ethanoic acid is the most polar and can **hydrogen-bond** to the silica, so it moves least. (b) A more polar solvent competes better for the compounds, so **all the Rf values increase**.

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5. One spot.

> **No.** Two substances can have the **same Rf** in one solvent. Run the sample again with a **different solvent** (or stationary phase), or use another method such as GC or a melting point; a pure sample gives one spot under every condition.

D. Gas chromatography

6. Retention times.

> (a) **Ethanol** and **butan-1-ol** (the retention times match the standards; no propan-1-ol).
> (b) Ethanol has the **lowest boiling point** (smallest molecule, weakest attractions), so it spends most time in the gas phase and leaves the column first.

7. Area percent.

> $\text{Total} = 2310 + 1540 + 3850 = 7700$
> $\dfrac{2310}{7700} \times 100\ \% = \mathbf{30.0\ \%}$; $\dfrac{1540}{7700} \times 100\ \% = \mathbf{20.0\ \%}$; $\dfrac{3850}{7700} \times 100\ \% = \mathbf{50.0\ \%}$

E. Replicates and quantities

8. Quintuplicate Rf.

> (a) Mean $= \dfrac{0.61 + 0.63 + 0.62 + 0.60 + 0.64}{5} = \mathbf{0.62}$; $s = \mathbf{0.016}$ (sample standard deviation, $n - 1 = 4$)
> (b) **Dye M.** Its Rf (0.62) equals the mean. Dye N (0.71) differs from the mean by 0.09, more than 5 standard deviations, so it does not match.

9. GC quantification.

> (a) Mean area $= \dfrac{1850 + 1870 + 1860}{3} = \mathbf{1860}$; $s = \mathbf{10}$; RSD $= \dfrac{10}{1860} \times 100\ \% = \mathbf{0.54\ \%}$
> (b) $c = 0.500\ \text{mg/mL} \times \dfrac{1860}{2480} = \mathbf{0.375\ mg/mL}$ (the areas cancel)
> (c) $m = 0.375\ \text{mg/mL} \times 20.0\ \text{mL} = \mathbf{7.50\ mg}$

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